Curriculum Vitæ *updated: November 5, 2026*



Philipp M. Schicho

1. Personal information

name	Dr. Philipp Maximilian Schicho born in Graz, Austria, 01 October 1991
nationality	Austrian
address	26, Quai Ernest-Ansermet CH-1211 Genève 4, Switzerland
email	philipp.schicho@unige.ch
website	pschicho.github.io
inspire HEP	inspirehep.net/authors/1639147
google scholar	scholar.google.com/citations?user=6BI62ioAAAAJ
ORCID iD	0000-0001-5869-7611
phone	+41 22 379 6314

2. Current position

Since 09/2024 — Research group leader (SNSF Ambizione fellow)
Thermal field theory, Particles, and Cosmic phases research group
Département de Physique Théorique, Université de Genève
Principal investigator of SNSF project no. 215997: *Gravitational waves from cosmological phase transitions: Precision cosmology in the light of LISA* (PRECLISA)

3. Employment history

09/2022 – 08/2024 — Postdoctoral researcher
Institute for Theoretical Physics, Goethe University Frankfurt

05/2020 – 08/2022 — Postdoctoral researcher
Helsinki Institute of Physics, University of Helsinki

02/2017 – 04/2020 — Doctor of Philosophy, PhD Physics (magna cum laude), 23/04/2020
AEC, Institute for Theoretical Physics, University of Bern
Advisor: M. Laine
Thesis: *Multi-loop investigations of strong interactions at high temperatures*, (cf. research output [4]).

10/2016 – 01/2017 Technical student
CERN, Accelerator and Beam Transfer, Beam Transfer Physics, (cf. research output [6]).

06/2015 – 08/2015 Summer student
CERN, Accelerator and Beam Transfer, Beam Transfer Physics
Thesis: *Optimising simulation times of SPS slow extraction using* `MAD-X`, (cf. research output [5]).

07/2014 – 08/2014 Summer student (GPA 1.0/1.0)
HEPHY, Institute of High Energy Physics, Vienna
Advisor: R. Schöfbeck
Thesis: *Increasing the sensitivity of a search for supersymmetry in the single lepton channel with the Stransverse Mass* M_{T2} (CMS), (cf. research output [1]).

4. Education

07/2017 École de physique des Houches
Effective Field Theory (EFT) in particle physics and cosmology

03/2017 Computer algebra and particle physics (CAPP) school, DESY, Hamburg

09/2014 – 11/2016 Master of Science, MSc Physics (GPA 5.5/6.0), 01/11/2016
ETH Zürich, Switzerland
Advisor: P. de Forcrand
Thesis: *Inhomogeneous condensation in quark-based QCD effective models via wavelet pseudoparticles*, (cf. research output [3]).

07/2014 LAPP Annecy-le-Vieux, France
Summer School in Particle and Astroparticle physics

08/2013 Theoretical Physics Summer school, University of Utrecht, Netherlands

09/2011 – 08/2014 Bachelor of Science, BSc Physics (with distinction, GPA 1.1/1.0), 12/08/2014
Graz University of Technology, Austria
Advisors: H. G. Evertz, C. B. Lang
Thesis: π- *and* ρ*-Meson mass spectroscopy from Lattice QCD*, (cf. research output [2]).

09/2002 – 05/2010 Österreichische Reifeprüfung, Matura (with distinction, GPA 1.0/1.0)
AHS BG/BRG Leibnitz, Austria
Thesis: *Sonoluminescence – A bubble's enlightenment.* A theoretical and experimental approach to the effect of Sonoluminescence.

5. Organization of conferences/symposia/workshops/seminars

09/2027 Conference organizer, *Phase Transitions from Microscopic to Cosmological Scales*, MITP, Mainz

06/2027 Conference organizer, *Synergies in non-equilibrium QFT: from heavy ions to the cosmos*, Bernoulli center, EPFL Lausanne

06/2027 Conference organizer, *Exploring Cosmological Phase Transitions with Gravitational Waves*, Nordita

01/2027 Main conference organizer, *CosmoFondue 2027*, CERN

08/2026 Conference organizer, *Strong and Electro-Weak Matter 2026*, University of Helsinki

05/2026 Workshop organizer, *Bubble Nucleation with Loop and Gravity Effects: Refining Gravitational-Waves Predictions*, IFPU (Trieste)

Since 2025 Seminar organizer, Cosmology and particle physics seminars, University of Geneva

08/2025 Workshop organizer, *Advancing gravitational wave predictions from cosmological first-order phase transitions*, CERN

08/2024 Conference organizer, *Strong and Electro-Weak Matter 2024*, Goethe University Frankfurt

06/2024 Convener, *2nd Edition of the Workshop 'Quarkonia meet Dark Matter' (QMDM)*, Institute for Advanced Study, Technical University of Munich

01/2024 Conference organizer, *Mainz-Frankfurt-Münster CosmoCoffee*, Mainz

6. Collaboration and outreach achievements

6.1. Collaboration between academia, industry and other actors in society

I also collaborate beyond physics. For the ophthalmology study [24], I conducted the statistical analysis and contributed to interpreting the results.

6.2. Collaboration with school and liberal adult education

09/2017 *Night of research* (*Nacht der Forschung* (NdF)), University of Bern
I presented research to a broad audience including school students and the general public.

6.3. Leadership experience and formal qualifications

Since 2026 *CERN guide* for the science gateway, antimatter factory, and synchrocyclotron. This role involves guiding visitors and communicating complex scientific concepts.

7. International research stays

06/2023 CERN, Genève, short-term visitor

12/2022 Kavli IPMU, Tokyo; visitor and plenary speaker

08/2018 Universidad del Bío-Bío, Grupo de Cosmología y Partículas Elementales; collaboration with Y. Schröder

8. Professional services and membership

Since 2022 Referee
American Physical Society's journals: Phys. Rev. **D**, Phys. Rev. **Lett.**, APS Open Science
Springer's journals: JHEP, Eur. Phys. J. **C**
Elsevier's journals: Comput. Phys. Commun.
Physics Essays

Since 2025 Core member of the LISA Cosmology Working Group

9. Scientific research skills

Theoretical (Dimensionally reduced) effective field theories, thermal field theory, quantum field theory, Lattice QCD, simulations in physics, computer algebra techniques, general relativity, cosmology, string theory, conformal field theory, group theory

Computational `C/C++`, `Python`, `Matlab`, `FORM`, `ROOT`, `FORTRAN 77`, Unix, Linux, Mathematica, LaTeX, computer hardware, `HTML`, Office, CAD-Software

10. Acquired funds

10.1. Third-party funding

2024 – 2028 Swiss National Science Foundation (SNSF) Ambizione Grant **215997**
Own function: Project manager and sole submitter
Overall funding: **CHF 808,885**

06/2027 Bernoulli center program funding for *Synergies in non-equilibrium QFT: from heavy ions to the cosmos*
Overall funding: **CHF 81,600**

06/2027 Nordita program funding for *Exploring Cosmological Phase Transitions with Gravitational Waves*
Overall funding: **SEK 540,000** (**CHF 45,000**)

01/2027 CERN TH conference funding for *CosmoFondue 2027*
Overall funding: **CHF 25,000**

05/2026 IFPU workshop funding for *Bubble Nucleation with Loop and Gravity Effects: Refining Gravitational-Waves Predictions*
Overall funding: **EUR 2,000**

08/2025 CERN TH institute funding for *Advancing gravitational wave predictions from cosmological first-order phase transitions*
Overall funding: **CHF 25,000**

10.2. Scholarships and individual funding

2011 – 2014 Scholarship of excellence Graz University of Technology
Overall funding: **EUR 2,400**

10.3. Prizes

06/2025 Awarded best poster prize at *CosmoFondue 2025*

11. Languages

German	Mother tongue
English	Proficient C2, TOEFL 106/120 (2014), Cambridge ESOL B2 First FCE (2010)
Spanish	Intermediate B1
Danish	Elementary A2
French	Beginner A2
Latin	Very good (literal translation)

12. Major scientific achievements

12.1. Precision thermodynamics for cosmological phase transitions

Invigorated by the first gravitational wave (GW) detections from binary mergers, I significantly improved the precision of thermodynamic computations relevant for cosmological phase transitions. I assessed theoretical uncertainties affecting such transitions in [11]. This work marks a **major step forward in the field of GW predictions from first-order phase transitions** with $\mathcal{O}(50)$ citations per year.

The reason for this leap in precision was that leading-order (LO) thermodynamic calculations are insufficient, rendering GW predictions ambiguous. Even minor uncertainties in the early stages of the beyond the Standard Model (BSM) $\rightarrow$ GW pipeline can drastically alter predictions, compromising stochastic gravitational wave background (SGWB) observations at LISA before they begin.

The work in [27] established the current state of the art in precision for GW predictions by completing perturbative computations for cosmological phase transitions. This accuracy was achieved by applying dimensional reduction to derive the three-dimensional SM EFT and by performing a three-loop computation of the effective potential. This framework has since been extended to full next-to-next-to-leading order in the hard thermal contributions [34].

12.2. Development of automated 3d EFT framework for thermal field theories

To improve the conceptual understanding of thermal EFT computations, I presented a didactic review [10] of thermal resummation techniques, combining perturbative and non-perturbative methods. Focusing on the dynamical real singlet extension of the Standard Model, a well-motivated BSM scenario for a strong first-order electroweak phase transition, I computed novel next-to-leading order (NLO) contributions to the parameters of the dimensionally reduced EFT. These results provided the foundation for subsequent non-perturbative lattice studies, which determined the phase structure of the model with unprecedented precision.

I demonstrated that two-loop corrections in both matching and effective potential are substantial [37, 26, 12] and essential for robust determinations of thermodynamic phase transition parameters. I also devised a minimal, gauge-invariant setup [16] that incorporates thermal resummation and supersedes previously used schemes.



In this context, I built a fully automated NLO pipeline that matches 4d thermal field theories to their 3d EFTs. This became the `Mathematica` package `DRalgo` [20], which is already used beyond QCD to deliver the **most accurate predictions of gravitational waves** from cosmological first-order phase transitions in generic BSM theories. With its most recent update [35], `DRalgo` is now capable of performing **dimension-six operator matching at one loop** for generic BSM theories, operators that can sizably shift GW predictions [30].

12.3. Gauge-invariant framework for nucleation at finite temperature

Theoretical uncertainties in determining the nucleation rate of bubbles of the stable vacuum at finite temperature remain a major challenge. In particular, radiatively induced transitions have long suffered from gauge-dependent ambiguities. I resolved this long-standing problem by employing the effective field theory at the nucleation scale for the Abelian Higgs model. This allowed me to demonstrate, for the first time [17, 14], gauge invariance of the LO perturbative contributions. Recently, I extended this framework to transitions with shifting scale hierarchies [37, 32, 28].

12.4. Model-generic determination of the bubble wall velocity during first-order phase transitions



I co-led the development of the public software package `WallGo` [29], which has become a major milestone for precise GW predictions. A key source of theoretical uncertainty in these predictions is the bubble wall velocity v_w, which crucially determines the resulting GW signal. For `WallGo`, I built and validated the code that solves a coupled system of hydrodynamic equations, classical scalar equations of motion, and Boltzmann equations, including out-of-equilibrium effects, to determine the bubble wall velocity v_w for generic models. **`WallGo` is the first public tool capable of computing v_w with full inclusion of out-of-equilibrium contributions**. By applying `WallGo`, I produced the first estimates of theoretical uncertainties in v_w from commonly used approximations in the literature, as demonstrated in [31].

13. List of publications

I have authored 37 publications to date, including 28 peer-reviewed international journal articles (1 of which is a co-authored review) and 3 conference proceedings. Altogether these works have accrued over 1800 citations (inspirehep.net, as of November 5, 2026), with a corresponding h-index of 20. Unless noted otherwise, the list below is ordered alphabetically. For each item I contributed at first- or second-author level: I carried out the computations, co-led the writing, and developed the main ideas. The complete publication list can also be found at Inspire HEP inspirehep.net/authors/1639147.

13.1. Journal articles

[41] F. Bernardo, M. Kierkla, N. Leister, **P. Schicho**, and P. Schwaller, *Bubble nucleation with thermal higher-dimensional operators*, (2026), [2610.12284].

[40] C. Benso, D. Bödeker, K. Kamada, K. Mukaida, L. Sagunski, **P. Schicho**, and K. Schmitz, *Baryon number freeze-out in the Standard Model, precisely*, (2026), [2609.07467].

[39] J. Chakrabortty, B. Siqueira Eduardo, S. Karmakar, **P. Schicho**, *Finite-temperature operator basis on $\mathbb{R}^3 \times S^1$ for SMEFT*, JHEP **09**, 241 (2026), [2605.02878].

[38] S. Biondini, **P. Schicho**, *A critical look at low-scale cosmological phase transitions in the PTA era*, (2026), [2607.02505].

[37] M. Kierkla, M. Lewicki, **P. Schicho**, D. Schmitt, and B. Swiezewska, *Theoretical uncertainties in reconstructing model parameters with gravitational waves from supercooled phase transitions*, (2026), [2607.18233].

[36] S. Bandyopadhyay, J. Chakrabortty, D. Dey, **P. Schicho**, and Tushar, *Higher-dimensional operators and Polyakov loop in hot Scalar QED from the heat kernel*, (2026), [2606.09779].

[35] F. Bernardo, R. G. Reinle, **P. Schicho**, *Matching higher-dimensional operators at finite temperature for general models*, (2026), [2605.15176].

[34] F. Bernardo, M. Chala, L. Gil, **P. Schicho**, *Hard thermal contributions to phase transition observables at NNLO*, JHEP **06**, 281 (2026), [2602.06962].

[33] A. Bhatnagar, D. Croon, **P. Schicho**, *Interpreting the 95 GeV resonance in the Two Higgs Doublet Model. Implications for the electroweak phase transition*, JHEP **03**, 014 (2026), [2506.20716].

[32] M. Kierkla, N. Ramberg, **P. Schicho**, and D. Schmitt, *Thermodynamical uncertainties for primordial black holes from cosmological phase transitions*, Phys. Rev. D **113**, 095024 (2026), [2506.15496].

[31] J. van de Vis, **P. Schicho**, L. Niemi, B. Laurent, J. Hirvonen, and O. Gould, *WallGo investigates: Theoretical uncertainties in the bubble wall velocity*, JHEP **04**, 041 (2026), [2510.27691].

[30] F. Bernardo, P. Klose, **P. Schicho**, and T. V. I. Tenkanen, *Higher-dimensional operators at finite temperature affect gravitational-wave predictions*, JHEP **08**, 109 (2025), [2503.18904].

[29] A. Ekstedt, O. Gould, J. Hirvonen, B. Laurent, L. Niemi, **P. Schicho**, and J. van de Vis, *How fast does the WallGo? A package for computing wall velocities in first-order phase transitions*, JHEP **04**, 101 (2025), [2411.04970].

[28] M. Kierkla, **P. Schicho**, B. Swiezewska, T. V. I. Tenkanen, and J. van de Vis, *Finite-temperature bubble nucleation with shifting scale hierarchies*, JHEP **07**, 153 (2025), [2503.13597].

[27] A. Ekstedt, **P. Schicho**, and T. V. I. Tenkanen, *Cosmological phase transitions at three loops: The final verdict on perturbation theory*, Phys. Rev. D **110**, 096006 (2024), [2405.18349].

[26] M. Lewicki, M. Merchand, L. Sagunski, **P. Schicho**, and D. Schmitt, *Impact of theoretical uncertainties on model parameter reconstruction from GW signals sourced by cosmological phase transitions*, Phys. Rev. D **110**, 023538 (2024), [2403.03769].

[25] J. Ghiglieri, **P. Schicho**, N. Schlusser, and E. Weitz, *The force-force correlator at the hard thermal scale of hot QCD*, JHEP **03**, 111 (2024), [2312.11731].

[24] C. Lwowski, K. Kaiser, J. Bucur, **P. Schicho**, and T. Kohnen, *Accuracy of using the axial length of the fellow eye for IOL calculation in retinal detachment eyes undergoing silicone oil removal*, Br. J. Ophthalmology (2023) 10.1136/bjo-2023-323581.

[23] J. Österman, **P. Schicho**, and A. Vuorinen, *Integrating by parts at finite density*, JHEP **08**, 212 (2023), [2304.05427].

[22] L. Sagunski, **P. Schicho**, and D. Schmitt, *Supercool exit: Gravitational waves from QCD-triggered conformal symmetry breaking*, Phys. Rev. D **107**, 123512 (2023), [2303.02450].

[21] G. Aarts, J. Aichelin, C. Allton, A. Athenodorou, D. Bachtis, C. Bonanno, N. Brambilla, E. Bratkovskaya, M. Bruno, M. Caselle, C. Conti, R. Contino, L. Cosmai, F. Cuteri, L. Del Debbio, M. D'Elia, P. Dimopoulos, F. Di Renzo, T. Galatyuk, J. N. Guenther, R. Houtz, F. Karsch, A. Y. Kotov, M. P. Lombardo, B. Lucini, L. Maio, M. Panero, J. M. Pawlowski, A. Pelissetto, O. Philipsen, A. Rago, C. Ratti, S. M. Ryan, F. Sannino, C. Sasaki, **P. Schicho**, C. Schmidt, S. Sharma, O. Soloveva, M. Sorba, and U.-J. Wiese, *Phase Transitions in Particle Physics: Results and Perspectives from Lattice Quantum Chromo-Dynamics*, Prog. Part. Nucl. Phys. **133**, 104070 (2023), [2301.04382].

[20] A. Ekstedt, **P. Schicho**, and T. V. I. Tenkanen, `DRalgo`*: A package for effective field theory approach for thermal phase transitions*, Comput. Phys. Commun. **288**, 108725 (2023), [2205.08815].

[19] T. Gorda, A. Kurkela, J. Österman, R. Paatelainen, S. Säppi, **P. Schicho**, K. Seppänen, and A. Vuorinen, *Degenerate fermionic matter at N3LO: Quantum electrodynamics*, Phys. Rev. D **107**, L031501 (2023), [2204.11893].

[18] T. Gorda, A. Kurkela, J. Österman, R. Paatelainen, S. Säppi, **P. Schicho**, K. Seppänen, and A. Vuorinen, *Soft photon propagation in a hot and dense medium to next-to-leading order*, Phys. Rev. D **107**, 036012 (2023), [2204.11279].

[17] J. Löfgren, M. J. Ramsey-Musolf, **P. Schicho**, and T. V. I. Tenkanen, *Nucleation at Finite Temperature: A Gauge-Invariant Perturbative Framework*, Phys. Rev. Lett. **130**, 251801 (2023), [2112.05472].

[16] **P. Schicho**, T. V. I. Tenkanen, and G. White, *Combining thermal resummation and gauge invariance for electroweak phase transition*, JHEP **11**, 047 (2022), [2203.04284].

[15] S. Biondini, **P. Schicho**, and T. V. I. Tenkanen, *Strong electroweak phase transition in t-channel simplified dark matter models*, JCAP **10**, 044 (2022), [2207.12207].

[14] J. Hirvonen, J. Löfgren, M. J. Ramsey-Musolf, **P. Schicho**, and T. V. I. Tenkanen, *Computing the gauge-invariant bubble nucleation rate in finite temperature effective field theory*, JHEP **07**, 135 (2022), [2112.08912].

[13] J. Ghiglieri, G. D. Moore, **P. Schicho**, and N. Schlusser, *The force-force-correlator in hot QCD perturbatively and from the lattice*, JHEP **02**, 58 (2022), [2112.01407].

[12] L. Niemi, **P. Schicho**, and T. V. I. Tenkanen, *Singlet-assisted electroweak phase transition at two loops*, Phys. Rev. D **103**, 115035 (2021), [2103.07467].

[11] D. Croon, O. Gould, **P. Schicho**, T. V. I. Tenkanen, and G. White, *Theoretical uncertainties for cosmological first-order phase transitions*, JHEP **04**, 055 (2021), [2009.10080].

[10] **P. M. Schicho**, T. V. I. Tenkanen, and J. Österman, *Robust approach to thermal resummation: Standard Model meets a singlet*, JHEP **06**, 130 (2021), [2102.11145].

[9] M. Laine, **P. Schicho**, and Y. Schröder, *A QCD Debye mass in a broad temperature range*, Phys. Rev. D **101**, 023532 (2020), [1911.09123].

[8] M. Laine, **P. Schicho**, and Y. Schröder, *Soft thermal contributions to 3-loop gauge coupling*, JHEP **2018**, 37 (2018), [1803.08689].

13.2. Conference proceedings

[7] J. Ghiglieri, G. D. Moore, **P. Schicho**, N. Schlusser, and E. Weitz. *Hard parton dispersion in the quark-gluon plasma, non-perturbatively.* In *11th International Conference on Hard and Electromagnetic Probes of High-Energy Nuclear Collisions* (July 2023). [2307.09297].

[6] M. Fraser, D. Björkman, K. Cornelis, B. Goddard, V. Kain, **P. Schicho**, C. Theis, and H. Vincke. *Modelling the Radioactivity Induced by Slow-Extraction Losses in the CERN SPS*. In *Proc. of International Particle Accelerator Conference (IPAC'17)* (May 2017), 1897–1900.

[5] M. A. Fraser, R. G. Alia, B. Balhan, H. Bartosik, C. Bertone, D. Björkman, J. Borburgh, N. Conan, K. Cornelis, L. Gatignon, B. Goddard, Y. Kadi, V. Kain, A. Mereghetti, F. Roncarolo, **P. M. Schicho**, J. Spanggaard, O. Stein, L. Stoel, F. M. Velotti, and H. Vincke. *SPS Slow Extraction Losses and Activation: Challenges and Possibilities for Improvement*. In *Proc. of International Particle Accelerator Conference (IPAC'17)* (Copenhagen. 2017), 611–614.

13.3. Theses

[4] **P. M. Schicho**, *Multi-loop investigations of strong interactions at high temperatures*, PhD thesis (U. Bern, 2020).

[3] **P. M. Schicho**, *Inhomogeneous condensation in quark-based QCD effective models via wavelet pseudoparticles*, MA thesis (ETH Zürich, 2016).

[2] **P. Schicho**, *π- and ρ-Meson mass spectroscopy from Lattice QCD*, BA thesis (TU Graz, 2014).

[1] **P. Schicho**, *Increasing the sensitivity of a search for supersymmetry in the single lepton channel with the Stransverse Mass*, Project thesis (HEPHY Vienna, 2014).

14. Presentations and conference contributions

I actively participate as a speaker at the leading international conferences and workshops in particle cosmology, thermal QCD, and phase transitions. Since 2017, I have delivered a total of 13 invited conference talks, 21 contributed conference talks, and 40 seminars.

14.1. Invited talks at conferences and workshops

[W13] *Fun with higher orders and operators*, invited **plenary** talk at *1st Granada workshop on phase transitions*, Granada, Spain, 15/10/2026.

[W12] *Flavor effects in baryon number freeze-out in the Standard Model, precisely*, invited **plenary** talk at *PPC 2026*, Sydney, Australia, 01/09/2026.

[W11] *Probing early universe phase transition microphysics with gravitational waves*, invited **plenary** talk at *ATHEXIS – Athens Symposium on Exploring the Universe*, Athens, Greece, 08/06/2026.

[W10] *From microphysics to Gravitational Waves for cosmological phase transitions*, invited **plenary** talk at *the 4th Topics of Particle, Astro and Cosmo Frontiers (TOPAC 2026)*, Chengdu, China, 21/04/2026.

[W9] *Construction and limitations of EFTs for hot QFT*, invited **plenary** talk at *EFT, everywhere, all at once*, CERN, Geneva, Switzerland, 20/03/2026.

[W8] *Cosmological phase transitions as sources of gravitational waves*, invited **plenary** talk at *Gravitational Waves and the Early Universe: Accelerated Expansion, Dynamical Inhomogeneity, and Beyond*, KMI, Nagoya University, Japan, 13/03/2026.

[W7] *Perturbative cosmological phase transitions in a broad temperature range*, invited **plenary** talk at *The 2025 Beijing Particle Physics and Cosmology Symposium (BPCS 2025)*, Beijing, China, 27/09/2025.

[W6] *Perturbative cosmological phase transitions in a broad temperature range*, invited **plenary** talk at *Numerical simulations of early Universe sources of gravitational waves*, NORDITA, Stockholm, Sweden, 04/08/2025.

[W5] *Limits and limitations of perturbative approaches to cosmological phase transitions*, invited talk at *GW day at CERN*, Geneva, Switzerland, 05/03/2025.

[W4] *Reliable theoretical predictions for GWs from first-order phase transitions*, invited **plenary** talk at *Fun with the phases of the Universe*, IFPU, Trieste, Italy, 08/10/2024.

[W3] *Reliable theoretical predictions for GWs from first-order phase transitions*, invited **plenary** talk at *Fundamental physics and gravitational wave detectors*, Pollica physics center, Italy, 20/09/2024.

[W2] *Strong cosmological phase transitions through the EFT lens*, invited **plenary** talk at *Quarkonia meet Dark Matter*, Institute for Advanced Study, Technical University of Munich, Germany, 19/03/2024.

[W1] *EFT framework for (precision) cosmological phase transition thermodynamics*, invited **plenary** talk at *What the heck happens when the Universe boils?*, Kavli IPMU, Tokyo, Japan, 05/12/2022.

14.2. Contributed talks

[C21] *Flavor effects in baryon number freeze-out in the Standard Model, precisely*, contributed talk at SEWM 2026, Helsinki, Finland, 20/08/2026.

[C20] *Thermodynamic limitations for primordial black holes from cosmological phase transitions*, contributed talk at NEHOP 2026 – New Horizons on Primordial Black Hole Physics, CERN, Switzerland, 18/06/2026.

[C19] *Higher-dimensional operators at finite temperature affect gravitational-wave predictions*, contributed talk at Extended Scalar Sectors From All Angles 2026, CERN, Switzerland, 04/06/2026.

[C18] *Uncertainty budgets for bubble wall velocities in cosmological phase transitions*, contributed talk at LISA Cosmology Working Group Workshop 2026, ICC, University of Barcelona, Spain, 03/06/2026.

[C17] *Uncertainty budgets for bubble wall velocities in cosmological phase transitions*, contributed talk at PLANCK2026 and 6th EuCAPT Symposium, CERN, Switzerland, 19/05/2026.

[C16] *Connecting micro- and intermediate scales in perturbative cosmological phase transitions*, contributed talk at Swiss LISA theory group meeting, UZH Zürich, Switzerland, 16/09/2025.

[C15] *Higher-dimensional operators at finite temperature affect gravitational-wave predictions*, contributed talk at Swiss Cosmology Days 2025, ETH Zürich, Switzerland, 06/06/2025.

[C14] *Higher-dimensional operators at finite temperature affect gravitational-wave predictions*, contributed talk at HEFT 2025, Geneva, Switzerland, 05/06/2025.

[C13] *Completing the perturbative program for cosmological phase transitions*, contributed talk at KEK theory meeting on particle physics phenomenology (KEK-PH2025winter), Tsukuba, Japan, 21/02/2025.

[C12] *Finite-temperature effective theories for first-order cosmological phase transitions*, contributed talk at High-Energy Theoretical Particle Physics Mini-Conference, Genève, Switzerland, 04/09/2024.

[C11] *Phase transition thermodynamic parameters at high precision*, contributed talk at 26th International Conference on Particle Physics and Cosmology (COSMO'23), Madrid, Spain, 14/09/2023.

[C10] *Degenerate fermionic matter at N^3LO*, contributed talk at ELEMENTS Annual Conference, Bad Nauheim, Germany, 11/07/2023.

[C9] *Can EFT reveal if there was an electroweak phase transition?*, contributed talk at

Cosmology from Home, 03/07/2023.

[C8] PRECLISA. *Gravitational waves from cosmological phase transitions: Precision cosmology in the light of LISA*, interview talk for the Ambizione grant of the Swiss National Science Foundation, Bern, Switzerland, 22/06/2023.

[C7] *Phase transition thermodynamic parameters at high precision*, contributed talk at How fast does the bubble grow?, DESY, Hamburg, Germany, 17/05/2023.

[C6] *Strong electroweak phase transition and simplified dark matter models*, contributed talk at Progress on Old and New Themes in cosmology (PONT) 2023, Avignon, France, 05/05/2023.

[C5] *Jet dispersion in hot QCD from the lattice*, contributed talk at SEWM 2022, Paris, France, 20/06/2022.

[C4] *(Non-)perturbative jet dispersion in hot QCD*, contributed talk at Quark Matter 2022, Kraków, Poland, 06/04/2022.

[C3] *(Non-)perturbative jet dispersion in hot QCD*, contributed talk at Mini workshop: Phase transitions in particle physics, Galileo Galilei Institute, Firenze, Italy, 30/03/2022.

[C2] *Soft thermal contributions to 3-loop gauge coupling*, contributed parallel talk at FunQCD (online), Barcelona, Spain, 29/03/2021.

[C1] *Fun with thermal dimension-six operators*, contributed parallel talk at SEWM 2018, Barcelona, Spain, 28/06/2018.

14.3. Seminar talks

[S40] *Precision thermal field theory for the early universe*, talk at the PT@Geneva meeting, CERN, Geneva, Switzerland, 05/11/2026.

[S39] *Baryon number freeze-out in the Standard Model, precisely*, invited seminar talk at Monash University, Australia, 23/09/2026.

[S38] *Finite-temperature bubble nucleation with shifting scale hierarchies*, invited seminar talk at University Melbourne, Australia, 21/09/2026.

[S37] *Finite-temperature bubble nucleation with shifting scale hierarchies*, invited seminar talk at University of Bologna, Italy, 14/05/2026.

[S36] *Finite-temperature bubble nucleation with shifting scale hierarchies*, invited seminar talk at Tor Vergata, Rome, Italy, 11/05/2026.

[S35] *Potentials, rates, and bubble velocities in cosmological phase transitions*, invited seminar talk at SISSA, Trieste, Italy, 04/05/2026.

[S34] *Cosmological phase transitions across all length scales*, invited seminar talk at UESTC, Chengdu, China, 22/04/2026.

[S33] *Microscopic descriptions of cosmological first-order phase transitions*, invited seminar talk at University of Stavanger, Norway, 09/04/2026.

[S32] *From microphysics to Gravitational Waves for cosmological phase transitions*, invited seminar talk at University of Oslo, Norway, 07/04/2026.

[S31] *Flavor effects on the baryon number freeze-out in the Standard Model, precisely*, invited seminar talk at Kobe University, Japan, 06/03/2026.

[S30] *Baryon number freeze-out in the Standard Model, precisely*, invited seminar talk at CERN, Geneva, Switzerland, 18/02/2026.

[S29] *Uncertainty budgets for bubble wall velocities in cosmological phase transitions*, invited seminar talk at Marietta Blau Institute for Particle Physics ÖAW, Vienna, Austria, 20/11/2025.

[S28] *Uncertainty budgets for bubble wall velocities in cosmological phase transitions*, invited seminar talk at ITP CAS, Beijing, China, 23/09/2025.

[S27] *Limits and limitations of perturbative approaches to cosmological phase transitions*, invited seminar talk at Warsaw University, Poland, 27/03/2025.

[S26] *Perturbative cosmological phase transitions and their bubble wall velocities*, invited seminar talk at Kavli IPMU, Tokyo, Japan, 26/02/2025.

[S25] *Completing the perturbative program for cosmological phase transitions*, invited seminar talk at Bielefeld University, Germany, 10/06/2024.

[S24] *Impact of computational diligence on GW signals from phase transitions*, invited seminar talk at Warsaw University, Poland, 23/11/2023.

[S23] *Impact of computational diligence on GW signals from phase transitions*, invited seminar talk at University of Basel, Basel, Switzerland, 12/10/2023.

[S22] *Impact of computational diligence on GW signals from phase transitions*, invited seminar talk at Helsinki Institute of Physics, Helsinki, Finland, 19/09/2023.

[S21] *Electroweak phase diagram at finite lepton number density*, contributed talk at Thermal Field Theory meets Phenomenology, Uppsala, Sweden, 15/06/2023.

[S20] *High-temperature effective field theories: Precision phase transition thermodynamics*, seminar talk at HEP seminar, Université de Genève, Switzerland, 30/05/2023.

[S19] *Integrating by parts at finite density*, informal talk at the DMGW meeting, Goethe University, Frankfurt, Germany, 10/05/2023.

[S18] *Degenerate fermionic matter at N^3LO*, invited seminar talk at Gravitation and Cosmology seminar, Utrecht University, Netherlands, 25/01/2023.

[S17] *What can EFT tell us about the electroweak phase transition?*, seminar talk at CRC-TR211 meeting and Colloquium, Bielefeld University, Germany, 24/01/2023.

[S16] *EFT framework for cosmological phase transition thermodynamics*, seminar talk at the AstroCoffee, Goethe University, Frankfurt, Germany, 08/11/2022.

[S15] *Degenerate fermionic matter at N^3LO*, invited seminar talk (online) at S@INT seminar, INT, Seattle, USA, 13/10/2022.

[S14] *(Gauge independent) Bubble nucleation rate at finite temperature*, invited seminar talk at Jožef Stefan Institute, Ljubljana, Slovenia, 15/09/2022.

[S13] *Can EFT tell us if there was an electroweak phase transition?*, invited seminar talk at University of Graz, Graz, Austria, 24/08/2022.

[S12] *Soft light-cone observables from electrostatic QCD*, invited seminar talk (online) at the QCD theory seminar, 11/07/2022.

[S11] *Degenerate fermionic matter at N^3LO*, invited seminar talk at the Nuclear Physics Colloquium, Goethe University, Frankfurt, Germany, 07/07/2022.

[S10] *Can EFT tell us if there was an electroweak phase transition?*, invited seminar talk at SUBATECH, Nantes, France, 16/06/2022.

[S9] *Electroweak phase transition: Combining thermal resummation and gauge invariance*, invited seminar talk at NICPB, Tallinn University, Estonia, 31/05/2022.

[S8] *Combining thermal resummation and gauge invariance for electroweak phase transition*, invited seminar talk (online) at School of Physics and Astronomy, Monash University, Australia, 24/05/2022.

[S7] *Effective theory approach to cosmological phase transitions*, invited seminar talk at Instituto de Astrofísica de Canarias, La Laguna, Spain, 03/03/2022.

[S6] *Gauge independent bubble nucleation rate at finite temperature*, invited seminar talk at University of Basel, Basel, Switzerland, 28/10/2021.

[S5] *Cosmological phase transition: Robust thermal resummation*, invited seminar talk at University of Bern, Bern, Switzerland, 19/10/2021.

[S4] *Cosmological phase transition: Robust thermal resummation*, invited seminar talk (online) at KIAS, Seoul, South Korea, 13/05/2021.

[S3] *How to be precise at the electroweak scale at finite temperature*, invited seminar talk (online) at Kavli IPMU, Tokyo, Japan, 25/11/2020.

[S2] *3-Loop Gauge Coupling in Hot Yang-Mills*, invited seminar talk at Helsinki Institute of Physics, Helsinki, Finland, 13/08/2019.

[S1] *Fun with thermal dimension-six operators*, invited seminar talk at Universidad del Bío-Bío, Chillán, Chile, 28/08/2018.

15. Teaching activities

15.1. Lecturer

09/2026 **Lecture**
Gravitational waves from first-order phase transitions,
MPA Retreat 2026 at the Castle Ebernburg
3 hours, PhD level, English, no. of students: ∼30

02/2025 **Lecture**
Bubble wall velocities in cosmological phase transitions,
Yonsei-Konkuk-Sogang mini-workshop on first-order phase transitions
5 hours, PhD and Postdoctoral level, English, no. of students: ∼20

2023/24 **Lecture**
Theoretische Physik 1: Mathematische Ergänzungen,
15 weeks × 2 hours/week (30 hours), Bachelor, German, no. of students: ∼20

03/2022 **Exercises**
Phase transitions in the early universe,
Theoretical Aspects of Astroparticle Physics, Cosmology and Gravitation
3 hours, PhD and Postdoctoral level, English, no. of students: ∼30

15.2. Teaching assistant

2022/23 **Exercises**: *Astrophysics II*, **2 credits**
15 weeks × 2 hours/week (30 hours), Bachelor, German, no. of students: ∼30

2019/20 **Exercises**: *Statistical thermodynamics II,* **4 credits**
15 weeks × 2 hours/week (30 hours), Bachelor, German, no. of students: ∼20

2019 **Exercises**: *Specialist course: Introduction to BSM physics,* **4 credits**
15 weeks × 2 hours/week (30 hours), Master, English, no. of students: ∼20

2018/19 **Exercises**: *Exercises in theoretical physics,* **6 credits**
15 weeks × 3 hours/week (45 hours), Master, English, no. of students: ∼30

2018 **Exercises**: *Quantum mechanics II,* **4 credits**
15 weeks × 2 hours/week (30 hours), Bachelor, German, no. of students: ∼50

2017/18 **Exercises**: *Quantum mechanics I,* **4 credits**
15 weeks × 2 hours/week (30 hours), Bachelor, German, no. of students: ∼50

2017 **Exercises**: *Standard Model,* **5 credits**
15 weeks × 2 hours/week (30 hours), Master, English, no. of students: ∼30

2016 **Exercises**: *Physics II,* **6 credits**
15 weeks × 2 hours/week (30 hours), Bachelor, German, no. of students: ∼30

2015/16 **Exercises**: *Numerical methods,* **6 credits**
15 weeks × 2 hours/week (30 hours), Bachelor, German, no. of students: ∼30

2015 **Exercises**: *Numerical analysis I,* **6 credits**
15 weeks × 2 hours/week (30 hours), Bachelor, German, no. of students: ∼50

2014 **Exercises**: *Theoretical mechanics,* **5 credits**
15 weeks × 2 hours/week (30 hours), Bachelor, German, no. of students: ∼20

2013/14 **Laboratory**: *Physics laboratory II,* **6 credits**
15 weeks × 2 hours/week (30 hours), Bachelor, German, no. of students: ∼20

2013 **Laboratory**: *Physics laboratory I,* **6 credits**
15 weeks × 2 hours/week (30 hours), Bachelor, German, no. of students: ∼20

15.3. Experience as supervisor

15.3.1. PhD students as primary supervisor

Since 2024 F. Bernardo (U. Geneva)

15.3.2. PhD students as joint supervisor

Since 2027 S. Kämpf (U. Turku) co-supervised with R. Paatelainen and T.V.I. Tenkanen

15.3.3. Master students as primary supervisor

2025/26 R. G. Reinle, (ETH Zürich)
Thesis: *The Heat-Kernel and functional matching methods for finite-temperature effective field theory*

2025/26 S. Kämpf, (U. Geneva)
Thesis: *Soft and hard effects on bubble wall dynamics in cosmological phase transitions*

15.3.4. Master students as joint primary supervisor

2021/22 S. Vihko, (U. Helsinki) co-supervised with A. Vuorinen
Thesis: *EFT methods and calculational techniques in imaginary time formalism of thermal QCD.*

15.3.5. Bachelor students as primary or joint primary supervisor

2022/23 R. Baumann, (U. Frankfurt) co-supervised with L. Sagunski, D. Schmitt
Thesis: *Gravitational waves in the Abelian Higgs model*

15.3.6. Project theses as primary supervisor

2024/25 B. Adadi, (ETH Zürich)

15.4. Pedagogical development work

2023/24 Developed a new lecture course on *Theoretische Physik 1: Mathematische Ergänzungen* with script: `https://pschicho.github.io/lectures/metp1.pdf`

2022 Developed exercises for the course *Phase transitions in the early universe* with script: `https://agenda.infn.it/event/28760/contributions/164435/attachments/89264/120616/ggi22_sol.pdf`

15.5. Pedagogical leadership

Since 2026 **Mentor** for Austrian high-school students

Since 2025 **Mentor** for the LISA Early Career Scientist (LECS) mentorship program

2022 – 2024 **PhD mentor** of R. Diedrichs, (U. Frankfurt) supervised by L. Sagunski

2022 – 2024 **PhD mentor** of D. Schmitt, (U. Frankfurt) supervised by L. Sagunski

2020 – 2024 **PhD mentor** of J. Österman, (U. Helsinki) supervised by A. Vuorinen